

May 11, 2026 03:15 PM GMT

Global Technology

Agentic AI – The Surge Begins

Agentic AI is no longer emerging but already reshaping industries – a new class of digital workforce is driving results. This means a rewrite of the AI infra equation – one with far more CPU and memory LTA capacity than earlier AI assumptions suggested, and hence much higher growth opportunity.

The 'agentic economy' is scaling faster than initially anticipated. This shift is already showing up in valuations – since mid-April (see our recent report [Rise of the AI Agent – Global Implications](#)), CPU and memory stocks have reached new highs. Recent 1Q26 commentary from AMD, Arm and Intel points to a recent surge in high-core-count CPUs and orchestration-heavy workloads, as hyperscaler data center deployments increasingly require denser CPU infrastructure around GPU clusters. At the same time, reported 3- to 5-year memory LTAs are emerging as a critical structural shift for the industry, extending pricing visibility and reinforcing durability in the cycle.

What's changed? 1Q26 has been an agent AI breakout quarter with a surge in 'agent-first' strategies across many companies, moving from experimental pilot programs to full-scale enterprise deployment. With accelerating enterprise adoption, CPU capacity requirements rise sharply and more measurable as agentic workloads scale. Now, in 2Q26, the debate is no longer about the potential of agentic AI, but about the speed of its integration. We also revise our memory framework to reflect higher orchestration CPU demand and rising DRAM content per CPU, driving incremental ~74EB in our base case, and ~228EB in a bull case by 2030.

Upgrading our CPU/memory TAM: We revise our framework to reflect faster agentic AI adoption, higher concurrency and more instructions per token. Our base-case total server CPU TAM rises 25% to US\$125bn by 2030, driven by a US\$79bn orchestration CPU layer on top of US\$45bn host/cloud CPU TAM; our top-down bull case reaches US\$283bn as dedicated agentic CPU racks scale. Applying the same CPU-led framework to memory, this implies ~74EB of incremental DRAM demand in our base case and ~221EB in our bull case (or 1.7x to 4.9x larger than the entire 2026 DRAM market).

Stay full-stack: The AI data center not only needs more GPUs but a stronger host CPU layer that can keep accelerators, memory, networking and software agents synchronized. Our preferred exposure is unchanged across CPUs, DRAM, NAND/eSSD, HDD, ABF substrates, foundry, memory interface, BMC, sockets/connectors and semi-cap, where content growth and supply constraints capture the “second leg” of the AI infrastructure cycle ([Exhibit 1](#)).

MORGAN STANLEY & CO. INTERNATIONAL PLC+

Shawn Kim

Equity Analyst

Shawn.Kim@morganstanley.com

+44 20 7677-1018

Lee Simpson

Equity Analyst

Lee.Simpson@morganstanley.com

+44 20 7425-3378

Nigel van Putten

Equity Analyst

Nigel.Putten@morganstanley.com

+44 20 7425-2803

Amelia M Scicluna

Research Associate

Amelia.Scicluna@morganstanley.com

+44 20 7425-6694

Cindy Huang

Equity Analyst

Cindy.Huang@morganstanley.com

+44 20 7425-2915

MORGAN STANLEY ASIA LIMITED+

Duan Liu

Equity Analyst

Duan.Liu@morganstanley.com

+852 2239-7357

Exhibit 1 : Likely beneficiaries of agentic AI take-up

Global Exposure Across the Stack ? Names by exposure					
CPU	DRAM	NAND	HDD	FOUNDRY	IC-design
• NVDA • Intel • AMD • Arm	• Samsung • Hynix • Micron	• Kioxia • SanDisk	• Seagate • WDC • TDK	• TSMC	• GUC • Egis
PCB/Substrate/CCL & Materials			BMC, CPU & Memory interface		
• SEMCO • Unimicron • NYPCB • Ibiben • Nittobo • MEC			• Aspeed • Renesas • Montage • WPG • AP Memory		
MLCC & CPU socket		ODM	SPE		
• Murata • TDK • Yageo • FIT Hon Teng • Lotes		• Wiwynn • Hon Hai	• ASML • ASMi • AMAT • Besi • KLAC • Tokyo Electron • Ulvac • Wonik		

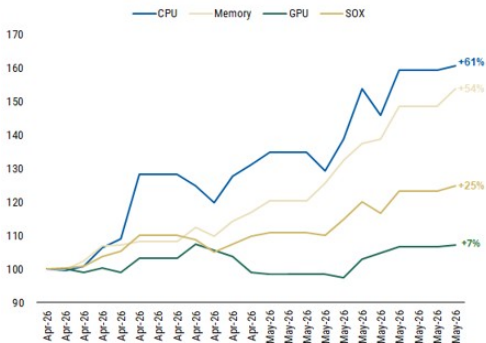
Source: Morgan Stanley Research

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+ = Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Exhibit 2 : CPU and memory have significantly outperformed the broader AI ecosystem since our last report



Source: Morgan Stanley Research. Note: CPU is an average performance of ARM, INTEL, AMD. Memory: Samsung, SK Hynix, Micron. GPU: NVIDIA

Agentic AI – Key Developments

We expect a surge in agentic AI competition to manifest between hyperscalers, frontier model companies, incumbent software vendors and new startups as companies race to build out money-making AI tools. But we are still early in the price discovery stage of the Agentic AI economy and still don't know yet what kind of unlock it is going to ultimately bring to drive the AI TAM, early in the structural shift in AI infrastructure from pure compute toward orchestration, memory and system-level coordination.

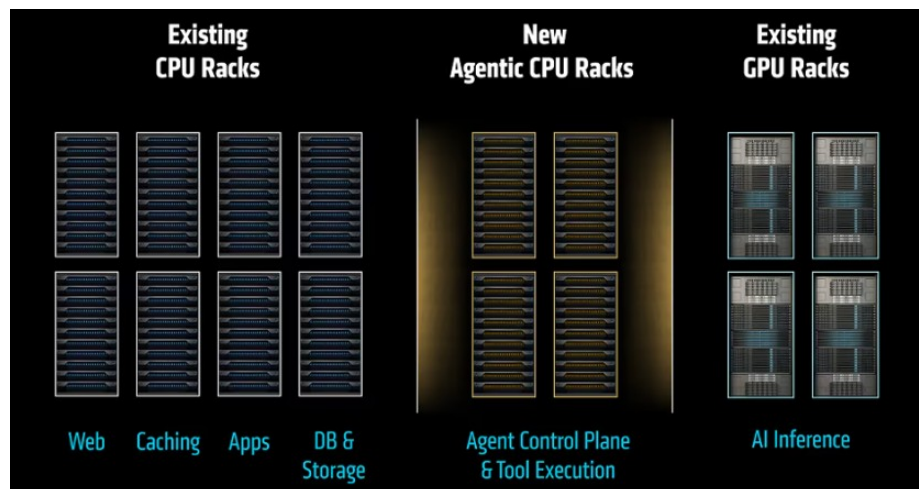
A pivot from search to action. We believe AI platforms are now shifting from cost centres to revenue infrastructure with the deployment of AI agents at scale in commerce, advertising and enterprise productivity. Agents drive more engagement, utility and customers stickiness on their platforms, due to their continuing learning ability and user context gained over time. Agents that conduct transactions becomes a significant value driver for hyperscale providers. Both Meta and Google have been reported to be working on AI agents, with Meta building a highly personalized AI assistant to carry out everyday tasks for its users (*Financial Times*, 6 May), and Google is developing a 24/7 personal agent for work, school and daily life, powered by Gemini (*Business Insider*, 6 May).

Agentic AI needs a rewrite of the AI infrastructure equation. Building an Agentic AI system does not mean putting a few more CPUs next to the same GPU-heavy rack design. It is a much more complex structural shift in data center architecture by adding a newly engineered CPU compute layer. Agentic AI requires entirely new racks of CPU servers that sit alongside GPU infrastructure and run to power the work of all these agents. The AI system in the future will look like a distributed system consisting of:

1. **GPU racks for dense model compute**, fast networking and a software stack that can keep it all observable, secure and efficient;
2. **Agentic CPU racks for orchestration**, processing data and tool execution.

In the agentic era, performance will not come from one processor doing everything but the right architecture – with CPUs and GPUs working together to move AI from answers to action. What will matter more for enterprises deploying AI agents is a balanced architecture, where: 1) the CPU tier is large enough to minimize GPUs wait, 2) efficient networking supporting agents, and 3) a data path allowing minimal latency. The orchestration layer is designed for concurrency to minimize cost and complexity.

Exhibit 3: Agentic AI infrastructure architecture – the CPU/GPU shift is far complex than adding more CPUs



Source: AMD, Morgan Stanley Research

A bigger TAM

We now expect the total addressable market for server CPUs and memory to grow meaningfully faster than our prior framework implied.

- We lift our base case total server CPU TAM to US\$125bn by 2030, from c.US \$100bn+ previously, as we account for agentic sessions, concurrent usage and instructions per token.
- In our bull case, using a top-down AI data center capacity framework, we estimate total server CPU TAM could reach US\$283bn by 2030.
- For memory, the same CPU-led framework implies a material uplift in DRAM demand, with agentic orchestration workloads driving ~74EB of incremental DRAM demand in our base case (or 1.7x 2026 total 45EB shipment) and ~221EB (4.9x this year's DRAM market) in our bull case by 2030.

Recent developments

AMD raised the server CPU TAM bar materially

AMD's 1Q26 call is the strongest CPU validation. Management now expects the server CPU TAM to grow >35% annually to >\$120bn by 2030, up from its prior ~18% CAGR framing, explicitly citing agentic AI as a driver of CPU compute for orchestration, data movement and parallel execution. AMD also reported Data Center revenue of \$5.8bn, +57% YoY, with server CPU revenue up >50% YoY and Q2 server CPU revenue expected to grow >70% YoY.

Arm's AGI CPU opportunity is scaling faster than expected

Arm reported record Q4 FYE26 revenue of \$1.49bn and full-year revenue of \$4.92bn, while data center royalties more than doubled YoY. More importantly, customer demand for the new Arm AGI CPU is now >\$2bn across FYE27-28, more than double what was indicated at Arm Everywhere. Arm also says its CPU compute share is around 50% among

top hyperscalers, with the AGI CPU developed with Meta and positioned for agentic AI infrastructure.

Intel validated the CPU / foundry / packaging angle

Intel's 1Q26 release explicitly said the shift from foundational models to inference to agentic AI is increasing demand for Intel's CPUs, wafer capacity and advanced packaging. Intel also reported DCAI revenue of \$5.1bn, +22% YoY, highlighted Xeon 6 as the host CPU for NVIDIA DGX Rubin NVL8, and noted Google's continued Xeon deployment plus a custom ASIC infrastructure processing unit collaboration.

Meta-AWS Graviton deal is the clearest real-world agentic CPU deployment

Meta signed an agreement with AWS to deploy tens of millions of Graviton cores for agentic AI workloads. AWS framed this as CPU-intensive demand from real-time reasoning, code generation, search and multi-step task orchestration, while Meta said agentic AI is evolving to require more CPU and that no single chip architecture can serve every workload.

Microsoft explicitly framed the cycle as the "agentic computing era"

Microsoft's FY3Q26 results used the phrase "agentic computing era", with revenue of \$82.9bn, +18% YoY. Management said it is focused on delivering cloud and AI infrastructure and solutions to help customers "eval-max" outcomes in the agentic computing era.

Alphabet / Google Cloud reinforced full-stack AI infrastructure demand

Google Cloud revenue grew 63% YoY to US\$20.0bn in 1Q26, supported by enterprise AI solutions and AI infrastructure, while the company positioned its new Gemini Enterprise Agent Platform as the connective layer for building, deploying and managing agents at scale. Google also launched an Agentic Data Cloud to help agents organize, understand and act on enterprise data in real time, committed US\$750mn to accelerate partner-led agentic AI development across its 120,000-member ecosystem, and introduced new TPU hardware (TPU 8t / 8i) designed for agentic workloads.

Memory LTAs are the most important structural change

Samsung and SK hynix have indicated they are moving toward 3-5 year LTAs with Big Tech customers. Memory and advanced packaging specifications for custom AI chips are increasingly locked in at the design stage, and that LTAs may include mechanisms such as price floors, supply volume commitments and binding agreements with very large up-front payments.

Base Case CPU TAM to US\$125bn

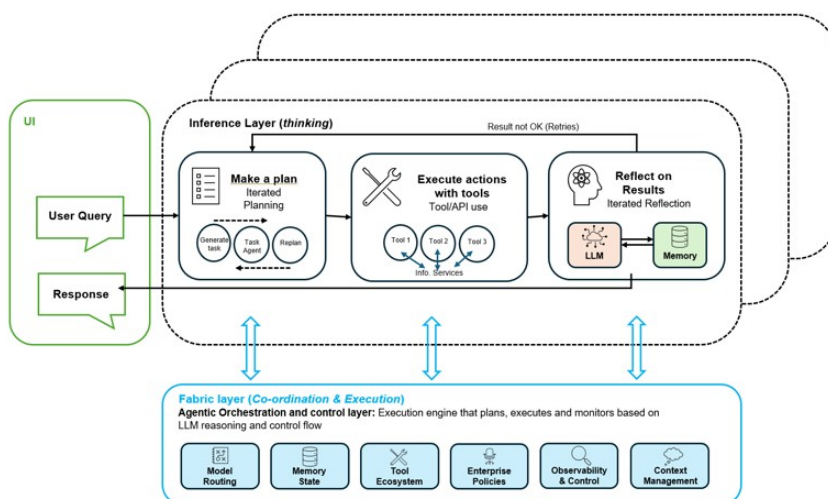
We revisit our CPU TAM model. As per our initial note, [Rise of the AI Agent - Global Implications](#), we maintain our bear case TAM of US\$77bn. However we now lift our base case to US\$125bn by 2030 as we account for agentic sessions, concurrent use and instructions per token. We also see a higher bull case which using a Top-Down model we estimate could reach US\$283bn.

A reminder of our thesis

We believe agentic AI will increase the CPU-to-GPU mix in AI systems by adding more orchestration, memory, and tool-use work around each model call. This should not reduce GPU demand, but it does increase overall system complexity and shifts incremental infrastructure spend toward CPUs, networking, and memory. In this environment, the advantage is less about owning the accelerator and more about owning the system architecture.

The initial GenAI wave was dominated by GPU-centric model-serving, with relatively light control-plane overhead. Agentic inference introduces multi-step workflows (plan, retrieve, call tools, execute, iterate), greater reliance on persistent memory and external tools/APIs, and higher orchestration complexity.

Exhibit 4: The inference fabric framework



Source: Morgan Stanley Research

From US\$100bn CPU TAM to US\$125bn

In our initial note, [Rise of the AI Agent - Global Implications](#), we parsed the Host CPUs from Orchestration CPUs. We used Mercury data to estimate the Host CPU market (Grace, Vera, Graviton etc) at US\$45bn by 2030. We continue to maintain this assumption in our new model. However, our previous estimates for the orchestration CPU TAM were conservative in particular our assumption for an AI semi TAM of US\$1.2Trn, note Nvidia are talking to this being as large as US\$3/4Trn. Our previous orchestration TAM of US\$30-

60bn implied a CPU TAM of US\$105bn, in-line with Arm's commentary at Arm Everywhere ([link](#)). However, AMD are now talking to a level closer to US\$120bn.

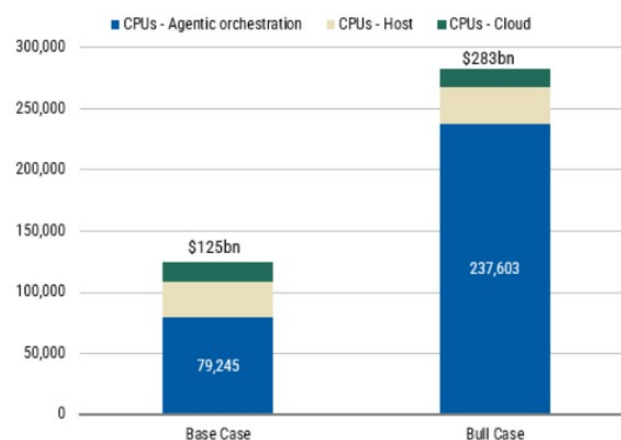
In deriving the Orchestration CPU TAM we take two different approaches: (i) a **bottom-up** model identifying agents per session, concurrent usage of compute and instructions per token; and (ii) a **top-down** model identifying GW installed per annum, implied racks and the ratio of CPU:GPU.

Exhibit 5: We lift our base case Orchestration CPU TAM to \$79bn (prior \$60bn).

TAM 2030 (\$bn)	Orch.	Host + cloud	Total
<i>Bear Case</i>	32	45	77
<i>Base (current) model</i>	60	45	105
<i>Base (new) Bottom-Up model</i>	79	45	125
<i>Bull Top-Down model</i>	238	45	283

Source: Morgan Stanley Research estimates

Exhibit 6: Our bull case implies a \$238bn CPU orchestration TAM.

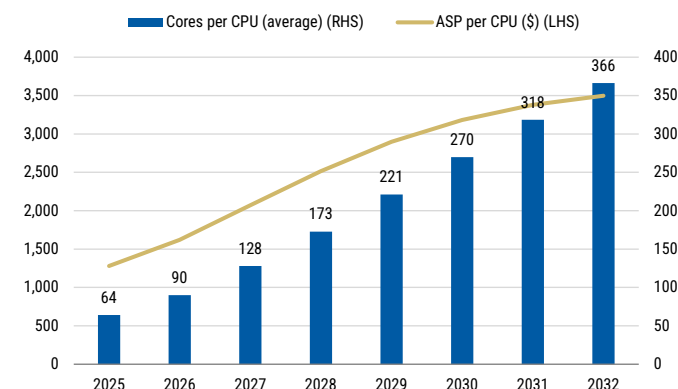


Source: Morgan Stanley Research estimates (e)

Assumptions underlying our model

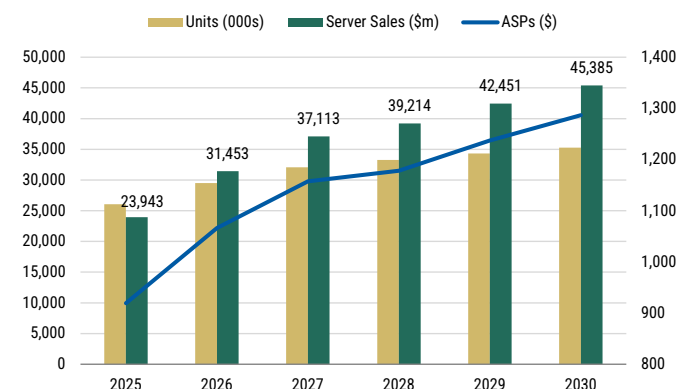
We assume in both our bottom-up and top-down model that a new generation of CPUs per annum implies that pricing per core declines 10% yy. However, in-line with current core trends we expect core counts per CPU to grow from 64 today to c.130 by FY27 and to 200-500+ by 2030. All in, this implies the FY27 ASP per CPU to come in at US \$2,000+ growing to at least US\$3,000 by 2030. For the head node market we model a combination of head node CPUs and regular CPU cloud servers. We expect CPU server sales to grow from US\$31bn FY26 to US\$45bn FY30.

Exhibit 7: We model the ASP per CPU to grow in-line with cores per CPU.



Source: Morgan Stanley Research estimates

Exhibit 8: We continue to expect CPU server sales to grow to \$45bn by 2030.

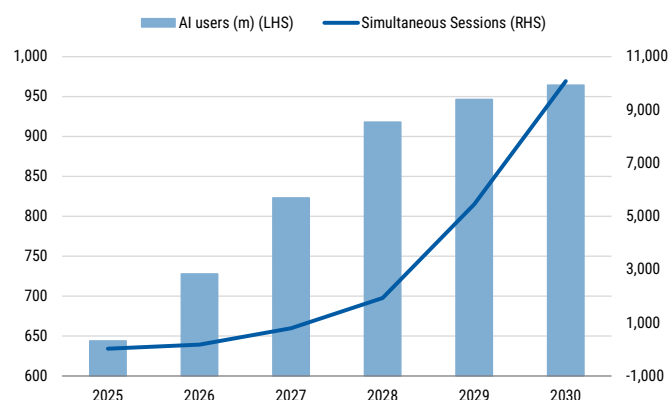


Source: Morgan Stanley Research estimates

A US\$79bn agentic CPU TAM base case...

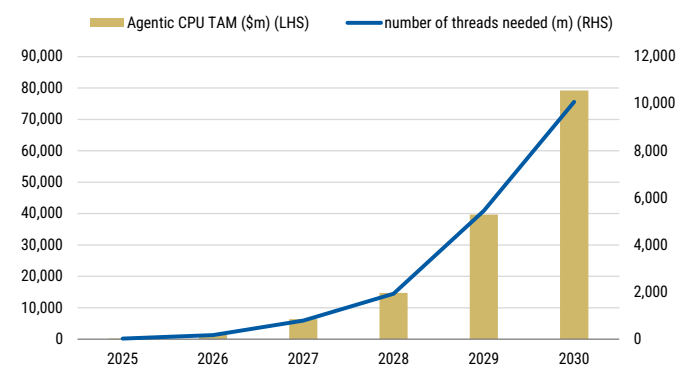
In deriving our base case we lean on the heavy use of agentic workflows by knowledge workers while assuming that single thread CPUs dominate. In this bottom-up model we estimate c.1bn knowledge workers globally by 2032 with the geographic mix as 50% Asia, c.25% Europe and 10 to 15% US. We estimate an AI take-up amongst knowledge workers of 78% for FY26 growing to 99% by 2030, while we model the occurrence of concurrent sessions to grow from 4% in FY26 to 19% in 2030. We assume from therein concurrency flattens at 15-20%. While 2 threads per core is the standard today, we see that reducing on the growing incidence of single threaded CPU cores amidst the need for power efficiency. We model the hyperthreading multiplier to decrease from 2 in 2026 to c.1.5 in 2030 implying the true availability of cores to grow at a 33% CAGR FY25 to FY29. On agentic use today we think today 6 agents per sessions is an accurate representation. However we expect this could almost reach 100 agents per session by 2032. We note this may still be conservative as exponential growth would imply trillions of agents per session. Assuming 1 session per thread, this implies 25mn CPUs by 2030. Applying our pricing assumptions this sets our base case orchestration TAM at US\$79bn FY30. When including our estimate for the Host and Cloud CPU we estimate a US\$125bn CPU TAM.

Exhibit 9: We model simultaneous sessions to grow at a 176% CAGR from FY26 to FY30.



Source: Morgan Stanley Research estimates

Exhibit 10: Our base case implies a \$79bn Agentic CPU TAM by 2030.

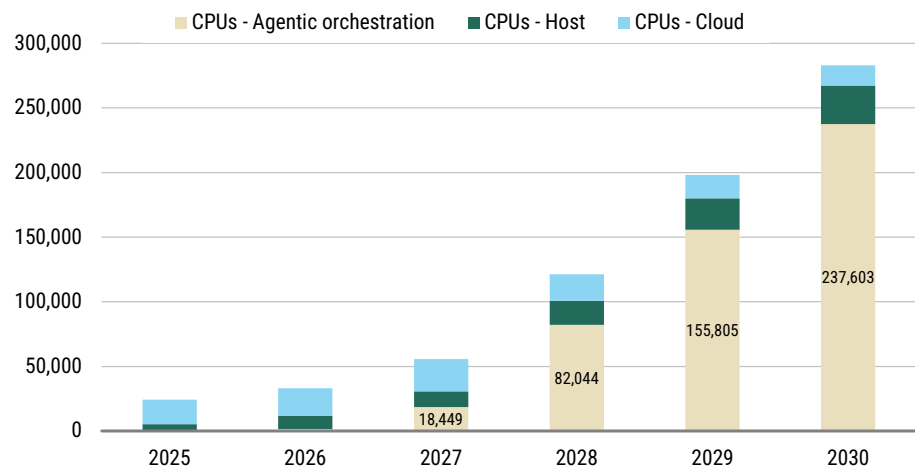


Source: Morgan Stanley Research estimates

... with our bull case implying a US\$238bn agentic CPU TAM.

To derive our bull case we look top-down at what AI data centre installed GW capacity may imply for the CPU:GPU ratio. Our Morgan Stanley global estimates imply AI data centre installed capacity grows from 24GW today before flattening to 35GW in the outer years. Using Nvidia racks as a market proxy we take the power budget per rack to calculate the GPU and CPU power draw per rack and what that would imply for the number of CPUs and GPUs per rack. Assuming pricing for Host CPUs are largely the same as orchestration CPUs this implies a \$25bn Host CPU TAM. We assume the CPU:GPU ratio ramps from 1:2 to 2:1 by 2030 implying orchestration CPUs grow at a 196% CAGR FY26 to FY30. This sets our orchestration CPU TAM at US\$238bn FY30. Accounting for host and cloud CPUs, our Bull Case CPU TAM is US\$283bn.

Exhibit 11: In our top-down model, we estimate the Agentic CPU TAM to grow at a 251% CAGR from FY26 to FY30.



Source: Morgan Stanley Research estimates

More reports @wechat: Hillwood2024

Raising Memory TAM

The revised CPU model materially raises the memory opportunity. Our prior framework estimated 15-45EB of incremental DRAM demand by 2030. We now estimate ~74EB in our base case and ~221EB in our bull case.

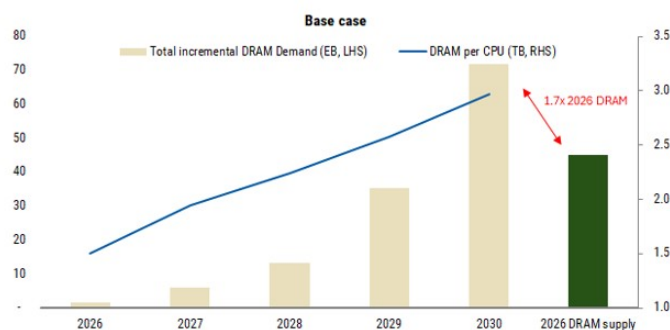
Agentic AI changes memory TAM in two ways: It raises the effective context per request and increases the amount of CPU-side memory provisioned per AI rack. As a result, the marginal memory demand shifts away from hot HBM alone and increasingly into host DRAM and rack SSD, where retained context, intermediate state and warm KV live closest to the CPU.

We keep the model intentionally simple:

Incremental DRAM demand = orchestration CPU units × DRAM per CPU

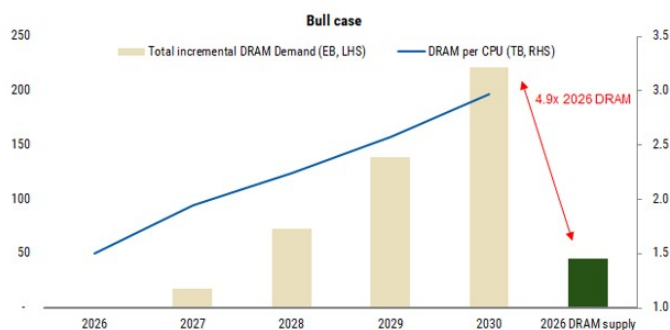
Our previous framework assumed 10-15mn incremental orchestration CPUs and 1.5-3.0TB DRAM per CPU, implying 15-45EB of incremental DRAM demand by 2030. The updated CPU TAM model points to a much larger orchestration CPU base. In our new base case, the US\$79bn orchestration CPU TAM implies roughly 25mn orchestration CPUs by 2030. Assuming DRAM per orchestration CPU rises toward 3TB as agent concurrency, memory services, RAG, KV-cache offload and persistent context scale, this implies roughly 74EB of incremental DRAM demand. In the bull case, the US\$244bn orchestration CPU TAM implies a much larger CPU pool and roughly 221EB of incremental DRAM demand by 2030.

Exhibit 12: Base case: agentic orchestration drives ~74EB incremental DRAM by 2030



Source: Morgan Stanley Research

Exhibit 13: Bull case: DRAM demand reaches ~221EB as orchestration CPU racks scale



Source: Morgan Stanley Research

Agentic AI – Global Beneficiaries

Exhibit 14: Agentic AI Beneficiaries

Company	Ticker	Rating	Last Close (LC)	Mkt Cap (\$ mn)	2026 E/L	2027 E/L	2026 E/L	2027 E/L	2026 E/L	2027 E/L	Reason
CPU											
Intel	INTC.O	EW	113	574,430	178.4	93.2	5.0	4.8	3%	6%	Incremental CPU demand
Nvidia	NVDA.O	OW	208	5,127,267	45.1	26.2	32.6	15.2	142%	122%	Well-positioned AI compute solution provider + Vera CPUs
AMD	AMD.O	EW	421	703,393	65.8	37.4	9.7	7.9	17%	26%	Incremental CPU demand
ARM	ARM.O	EW	237	243,812	94.8	121.0	19.2	23.2	25%	25%	Incremental CPU demand
HDD											
WDC	WDC.O	OW	483	187,888	48.5	24.3	18.0	15.9	68%	76%	A rising AI tides lifts all boats for both HDD and eSSD
Seagate	STX.O	OW	786	181,738	52.7	24.8	109.5	60.2	NM	427%	A rising AI tides lifts all boats for both HDD and eSSD
NAND											
Sandisk	SNDK.O	OW	1,410	203,742	23.0	8.3	12.2	5.1	103%	141%	NAND Supercycle supported by agentic AI + inference demand
Kioxia	285A.T	OW	43,410	153,527	12.1	11.7	7.4	4.6	156%	63%	NAND Supercycle supported by agentic AI + inference demand
DRAM											
Samsung Electronics	005930.KS	OW	271,500	1,345,222	5.6	4.1	2.7	1.7	86%	66%	Better memory cycle driven by AI + HBM market share gain optionality
SK hynix	000660.KS	OW	1,654,000	826,960	5.5	3.7	3.6	1.8	188%	97%	Better memory cycle driven by AI
Micron	MLU.O	OW	667	772,063	11.3	8.2	6.4	3.7	125%	76%	Better memory cycle driven by AI
ABF Substrates & Materials											
SDICO	009150.KS	OW	917,000	47,945	46.9	28.2	6.6	5.4	16%	23%	Rising demand for high value-added ABF package substrates
Unimicron	3037.TW	OW	896	42,972	91.4	45.0	11.6	9.7	14%	26%	Better server CPU demand to support higher volumes and more favorable pricing of ABF substrates
Nan Ya PCB	8046.TW	OW	947	19,507	113.0	63.3	12.2	10.9	12%	19%	Better networking ABF substrate demand driven by higher general server volumes
Ibiden	4062.T	OW	16,375	29,249	91.0	64.4	8.3	7.5	10%	13%	Rising demand for high value-added ABF package substrates
Nitrobo	3130.T	OW	31,950	7,438	62.2	38.9	6.4	5.7	14%	16%	Nitrobo supplies low-CTE glass cloth used in semiconductor package substrates
MEC	4971.T	OW	10,420	1,231	37.5	32.1	5.7	5.1	17%	18%	Larger ABF substrate sizes and a higher number of wiring layers bode well for MEC's C2 series adhesion promoters
Shennan	002916.SZ	EW	319	29,769	48.3	33.2	7.7	6.2	26%	32%	Benefit from increasing server, networking and optical module PCB demand
Shengyi	000183.SS	EW	81	27,097	39.2	28.4	9.0	7.6	26%	32%	Benefit from increasing server, networking and optical module PCB/OCL demand
Foundry											
TSMC	2330.TW	OW	2,310	1,908,987	22.4	17.9	8.0	5.9	49%	44%	Rising CPU TAM is increasingly captured by TSMC, supported by leading-edge node migration (2nm/3nm)
BMC/Memory Interface											
Aspeed	5274.TWO	OW	18,700	22,383	62.2	68.8	58.7	49.9	101%	97%	~70% of market shares in the CPU server BMC market
Marriage	6809.HK	OW	322	33,198	1.9	1.0	0.3	0.2	28%	30%	Higher CPU and DRAM intensity driving demand for memory interconnect solutions
Renesas	6723.T	OW	3,623	42,724	14.0	17.8	2.3	2.1	20%	13%	A beneficiary of the agentic AI shift through its exposure to CPU- and DRAM-facing memory interface
IP/Design Services											
QUC	3443.TW	OW	5,450	23,429	63.8	54.0	39.3	26.2	61%	73%	Strong demand for Arm-based CPUs drives incremental upside
Egis	6402.TWO	EW	129	375	NM	24.1	1.3	1.3	-1%	5%	Joining ARM's total design alliance with the aim of winning long-term ASIC business opportunities
Components/ODM											
Murata	6981.T	OW	5,648	67,511	35.3	28.8	3.9	3.7	11%	13%	Rising demand for highly durable and reliable electronic components/MLCC
TDK	6762.T	OW	2,896	35,140	23.7	21.1	2.6	2.4	12%	12%	Rising demand for highly durable and reliable electronic components/MLCC
Fit Hon Teng	6088.HK	OW	8	6,954	25.3	16.9	2.7	2.4	11%	16%	Clear beneficiaries of more CPU socket demand
Yageo	2327.TW	OW	372	23,645	22.3	19.4	3.6	3.3	20%	19%	Benefitting from increasing penetration of its products for both AI and general servers
Lotus	3533.TW	EW	2,850	9,519	27.4	23.4	6.7	5.7	29%	28%	CPU socket, PCIe/DRAM connector beneficiary
Gold Circuit Electronics	2366.TW	OW	1,450	23,876	34.8	22.2	14.6	10.1	61%	66%	Beneficiary of higher general server volumes and CPU MB PCB content growth
Wiwynn	6666.TW	OW	4,880	26,348	14.6	11.9	5.7	4.5	50%	48%	Key general server ODM
Lenovo	0992.HK	EW	12	18,634	8.4	11.2	1.9	2.2	26%	22%	General server supplier for select CSPs (ie. MSFT, ORCL)
Inspur	000977.SZ	EW	73	15,628	32.3	24.2	4.2	3.7	15%	17%	Clear ODM beneficiary of more AI and general server demand
SPE/Automation Equipment											
ASML	ASML.AS	OW	1,305	608,152	42.7	28.1	21.9	14.4	61%	77%	ASML benefits from increased EUV layer count
Best	BEES.AS	OW	255	23,701	69.3	45.1	20.6	15.6	42%	47%	Increasing accelerator demand drives greater adoption of advanced packaging technologies
Wainip PS	240810.KQ	OW	132,800	4,515	34.3	24.1	5.9	4.9	29%	24%	Strong rebound on the back of Samsung 12 DRAM capex ramp up
Applied Materials	AMAT.O	OW	429	345,059	37.1	27.9	13.4	10.8	45%	48%	The most leverage to greenfield DRAM SPE beneficiary under US SPE coverage
KLA Corp	KLAC.O	OW	1,816	239,296	50.5	35.3	40.0	29.3	101%	112%	Rising compute and CPU TAM drive incremental WFE demand
Tokyo Electron	8035.T	OW	51,720	152,191	32.9	21.1	9.6	7.8	34%	46%	A potential incremental share gainer on Intel's capex recovery
Ulvac	8025.T	OW	3,385	1,906	27.8	20.6	1.5	1.5	6%	7%	HMM tool has seen solid adoption at Intel
Bislink	3665.TW	OW	2,780	17,134	37.2	23.0	9.4	7.1	31%	40%	Semi cap equipment business (power interconnect and subsystem assembly business)
Hiwin	2048.TW	OW	318	3,586	38.0	31.0	2.1	2.0	8%	9%	Automation components
AirTac	1590.TW	OW	1,490	9,506	28.2	22.7	5.1	4.6	22%	23%	Automation components
Thermal/Power solution											
AVC	3017.TW	OW	2,415	28,945	25.5	18.7	11.8	8.1	73%	63%	Rising CPU/general server demand drives greater demand for advanced cooling solutions
Auras	3324.TWO	EW	1,105	3,039	19.0	16.7	6.9	5.5	47%	39%	Rising CPU/general server demand drives greater demand for advanced cooling solutions
Delta	2308.TW	OW	2,280	188,768	54.9	35.0	13.5	10.5	33%	42%	Elevated power consumption at data center level supports stronger growth for power supply players
Lite-On	2301.TW	EW	204	14,759	23.3	18.2	4.9	4.7	22%	27%	Elevated power consumption at data center level supports stronger growth for power supply players
EDA											
Synopsys	SNPS.O	EW	504	78,958	34.4	28.1	3.0	2.8	10%	11%	Benefitting from rising compute intensity and design complexity
Cadence	CDNS.O	OW	355	96,799	44.7	36.6	13.3	11.6	40%	36%	Benefitting from rising compute intensity and design complexity
Analog											
Infineon	IFXN.DE	OW	59	90,364	34.4	21.4	3.7	3.3	13%	17%	Potential upside to outer year data centre estimates from the development of agentic systems with multi-step workflows
STMicroelectronics	STM.PA	OW	49	51,511	48.8	21.5	2.8	2.5	6%	13%	Potential upside to outer year data centre estimates from the development of agentic systems with multi-step workflows

Source: Factset, Morgan Stanley Research

More reports @wechat: Hillwood2024

This report references U.S. Executive Order 14032 and/or entities or securities that are designated thereunder. U.S. persons may be prohibited from buying certain securities of entities named in this report. Readers are solely responsible for ensuring that their investment activities are carried out in compliance with applicable laws.

This report references export controls and/or entities that may be subject to export control restrictions. Readers are solely responsible for ensuring that their investment or trade activities are carried out in compliance with applicable laws.

This report references U.S. Executive Order 14105 and/or entities that may be in scope of such order. U.S. persons may be prohibited from engaging in certain transactions or otherwise require certain other transactions be notified to the U.S. Department of Treasury. Readers are solely responsible for ensuring that their investment or trade activities are carried out in compliance with applicable laws.

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley & Co. LLC and/or Morgan Stanley C.T.V.M. S.A. and/or Morgan Stanley México, Casa de Bolsa, S.A. de C.V. and/or Morgan Stanley Canada Limited and/or Morgan Stanley & Co. International plc and/or Morgan Stanley Europe S.E. and/or RMB Morgan Stanley Proprietary Limited and/or Morgan Stanley MUFG Securities Co., Ltd. and/or Morgan Stanley Capital Group Japan Co., Ltd. and/or Morgan Stanley Asia Limited and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105), Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Cindy Huang; Shawn Kim; Duan Liu; Nigel van Putten; Amelia M Scicluna; Lee Simpson.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

The analyst or strategist (or a household member) identified below owns the following securities (or related derivatives): Cindy Huang - ASML Holding NV (common or preferred stock).

As of April 30, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: Advanced Micro Devices, AirTAC International, AP Memory Technology Corp, Applied Materials Inc., Arm Holdings plc, Asia Vital Components Co. Ltd., ASML Holding NV, Aspeed Technology, Auras Technology Co Ltd, Bizlink, Cadence Design Systems Inc, Delta Electronics Inc., Global Unichip Corp, Gold Circuit Electronics Ltd., Hiwin Technologies Corp., Ibiden, Infineon Technologies AG, Inspur Electronic Information, Intel Corporation, KIOXIA Holdings, KLA Corp, MEC, Micron Technology Inc., Montage Technology Co Ltd, Nan Ya PCB, Nitto Boseki, NVIDIA Corp., Samsung Electro-Mechanics, SanDisk Corporation., Seagate Technology, SK hynix, STMicroelectronics NV, Synopsys Inc., TDK, Tokyo Electron, TSMC, Ulvac, Unimicron, Western Digital, Wiwynn Corp, WPG Holdings, Yageo Corp..

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Intel Corporation, KIOXIA Holdings, Montage Technology Co Ltd, Samsung Electronics, Seagate Technology, SK hynix, Wiwynn Corp.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Advanced Micro Devices, Applied Materials Inc., BE Semiconductor Industries NV, Intel Corporation, KIOXIA Holdings, Lenovo, Micron Technology Inc., Montage Technology Co Ltd, Seagate Technology, SK hynix, STMicroelectronics NV, Tokyo Electron, Wiwynn Corp, Yageo Corp..

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Advanced Micro Devices, AirTAC International, AP Memory Technology Corp, Applied Materials Inc., Arm Holdings plc, Asia Vital Components Co. Ltd., ASML Holding NV, BE Semiconductor Industries NV, Bizlink, Cadence Design Systems Inc, Delta Electronics Inc., FIT Hon Teng Ltd, Gold Circuit Electronics Ltd., Hon Hai Precision, Ibiden, Infineon Technologies AG, Intel Corporation, KIOXIA Holdings, KLA Corp, Lenovo, Lite-On Technology, MEC, Micron Technology Inc., Montage Technology Co Ltd, Murata Manufacturing, Nitto Boseki, NVIDIA Corp., Renesas Electronics, Samsung Electronics, SanDisk Corporation., Seagate Technology, SK hynix, STMicroelectronics NV, Synopsys Inc., TDK, Tokyo Electron, TSMC, Ulvac, Unimicron, Western Digital, Wiwynn Corp, WPG Holdings, Yageo Corp..

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Advanced Micro Devices, Applied Materials Inc., BE Semiconductor Industries NV, Cadence Design Systems Inc, Hon Hai Precision, Infineon Technologies AG, Intel Corporation, Lenovo, Micron Technology Inc., NVIDIA Corp., Seagate Technology, Synopsys Inc., Tokyo Electron, TSMC, Yageo Corp..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Advanced Micro Devices, AirTAC International, AP Memory Technology Corp, Applied Materials Inc., Arm Holdings plc, Asia Vital Components Co. Ltd., ASML Holding NV, BE Semiconductor Industries NV, Bizlink, Cadence Design Systems Inc, Delta Electronics Inc., FIT Hon Teng Ltd, Gold Circuit Electronics Ltd., Hon Hai Precision, Ibiden, Infineon Technologies AG, Intel Corporation, KIOXIA Holdings, KLA Corp, Lenovo, Lite-On Technology, Micron Technology Inc., Montage Technology Co Ltd, Murata Manufacturing, NVIDIA Corp., Renesas Electronics, Samsung Electronics, SanDisk Corporation., Seagate Technology, SK hynix, STMicroelectronics NV, Synopsys Inc., TDK, Tokyo Electron, TSMC, Ulvac, Unimicron, Western Digital, Wiwynn Corp, WPG Holdings, Yageo Corp..

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Advanced Micro Devices, Applied Materials Inc., BE Semiconductor Industries NV, Cadence Design Systems Inc, Hon Hai Precision, Infineon Technologies AG, Intel Corporation, Lenovo, Micron Technology Inc., Montage Technology Co Ltd, NVIDIA Corp., Renesas Electronics, Seagate Technology, Synopsys Inc., Tokyo Electron,

TSMC, Yageo Corp..

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of April 30, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1546	42%	467	51%	30%	709	44%
Equal-weight/Hold	1568	43%	358	39%	23%	715	44%
Not-Rated/Hold	4	0%	0	0%	0%	1	0%
Underweight/Sell	555	15%	84	9%	15%	202	12%
Total	3,673		909			1627	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O or Over) - The stock's total return is expected to exceed the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Not-Rated (NR) - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U or Under) - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

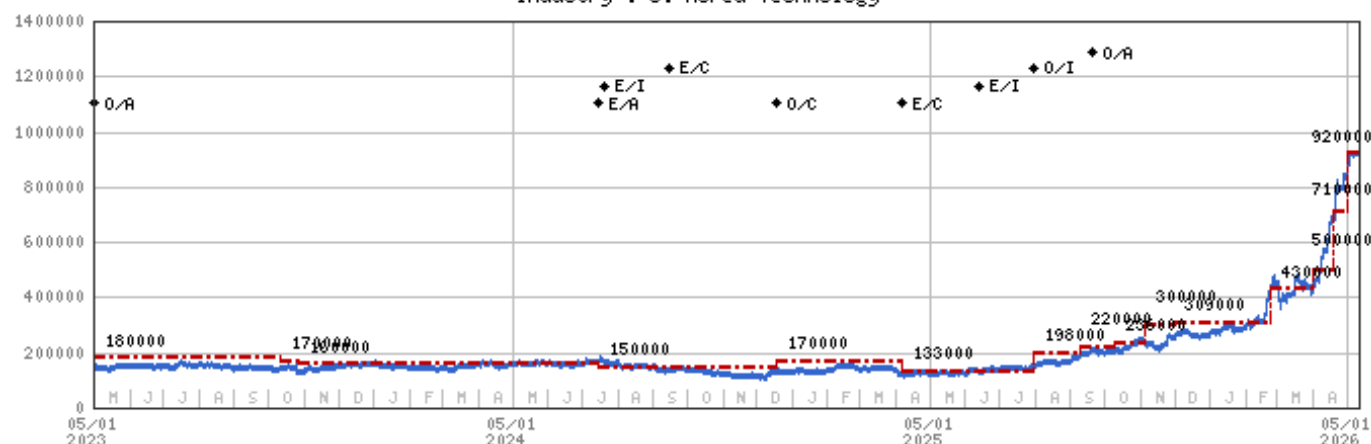
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Samsung Electro-Mechanics (009150.KS) - As of 05/11/26 GMT in KRW
Industry : S. Korea Technology



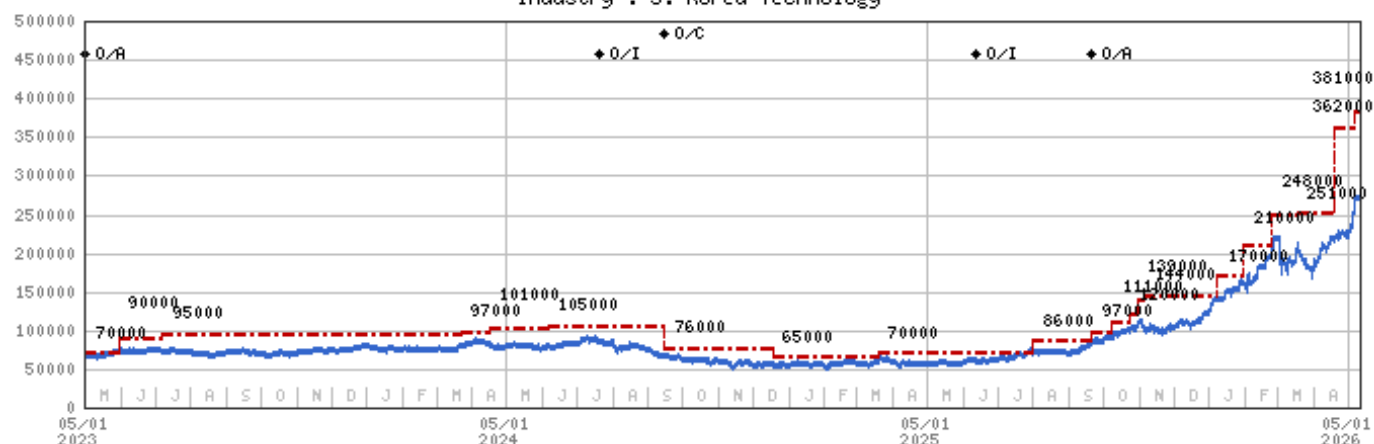
Stock Rating History: 5/1/21 : O/A; 5/12/21 : E/A; 7/19/21 : E/I; 8/12/21 : E/C; 6/1/22 : U/C; 10/4/22 : U/A; 11/9/22 : O/A; 7/15/24 : E/A; 7/21/24 : E/I; 9/15/24 : E/C; 12/18/24 : O/C; 4/7/25 : E/C; 6/13/25 : E/I; 7/31/25 : O/I; 9/21/25 : O/A
Price Target History: 3/1/21 : 250000; 5/12/21 : 180000; 10/19/21 : 170000; 10/27/21 : 160000; 1/24/22 : 180000; 3/18/22 : 170000; 6/1/22 : 140000; 11/9/22 : 180000; 10/11/23 : 170000; 10/27/23 : 160000; 7/15/24 : 150000; 12/18/24 : 170000; 4/7/25 : 133000; 7/31/25 : 198000; 9/10/25 : 220000; 10/10/25 : 236000; 11/5/25 : 300000; 12/1/25 : 309000; 2/23/26 : 430000; 4/2/26 : 500000; 4/19/26 : 710000; 5/1/26 : 920000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Samsung Electronics (005930.KS) - As of 05/11/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 5/1/21 : O/A; 7/19/21 : O/I; 8/12/21 : O/C; 10/4/22 : O/A; 7/21/24 : O/I; 9/15/24 : O/C; 6/13/25 : O/I; 9/21/25 : O/A

Price Target History: 2/25/21 : 115000; 5/18/21 : 93000; 6/8/21 : 98000; 8/12/21 : 89000; 9/15/21 : 95000; 12/3/21 : 97000; 3/18/22 : 95000; 4/28/22 : 85000; 6/10/22 : 80000; 7/5/22 : 75000; 7/22/22 : 70000; 9/17/22 : 68000; 3/21/23 : 70000; 5/30/23 : 90000; 7/7/23 : 95000; 3/22/24 : 97000; 4/16/24 : 101000; 6/6/24 : 105000; 9/15/24 : 76000; 12/18/24 : 65000; 3/19/25 : 70000; 8/1/25 : 86000; 9/21/25 : 97000; 10/8/25 : 111000; 10/23/25 : 120000; 10/30/25 : 139000; 11/5/25 : 144000; 1/8/26 : 170000; 1/30/26 : 210000; 2/24/26 : 248000; 3/18/26 : 251000; 4/19/26 : 362000; 5/6/26 : 381000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

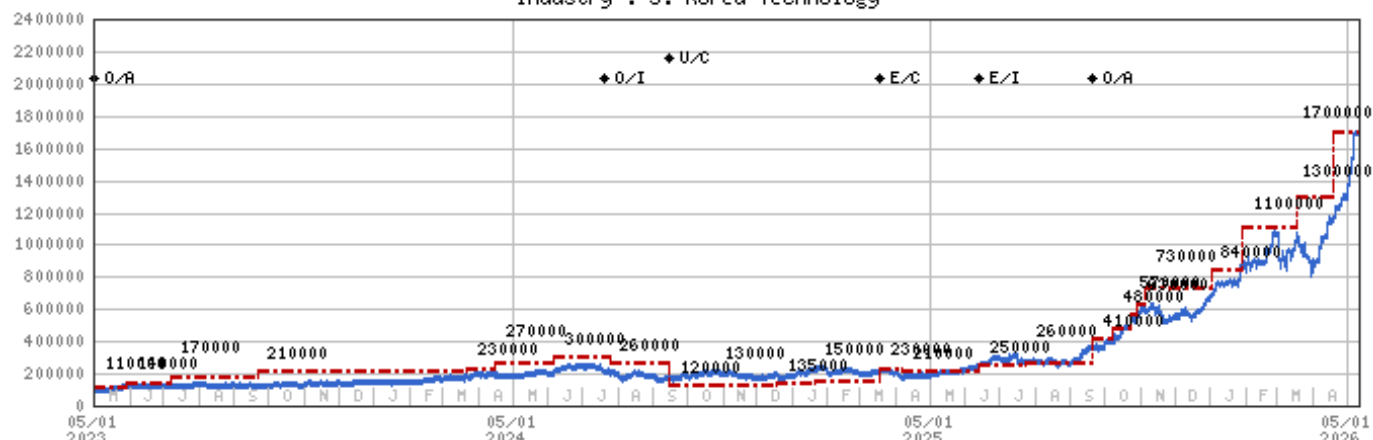
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

SK hynix (000660.KS) - As of 05/11/26 GMT in KRW
Industry : S. Korea Technology

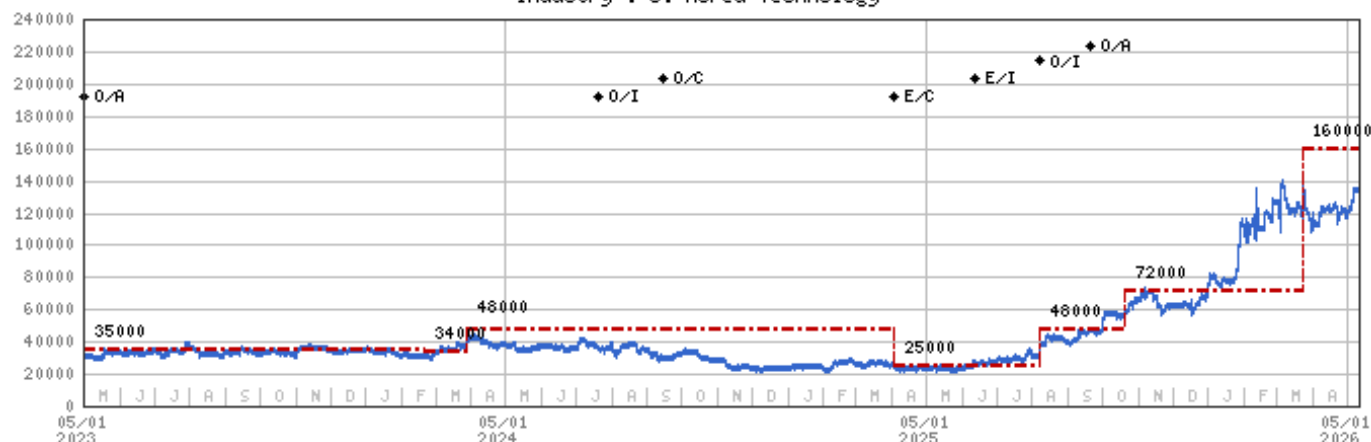


Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Wonik IPS Co Ltd (240810.KQ) - As of 05/11/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 5/1/21 : O/A; 7/19/21 : O/I; 8/12/21 : O/C; 10/4/22 : O/A; 7/21/24 : O/I; 9/15/24 : O/C; 4/3/25 : E/C; 6/13/25 : E/I; 8/8/25 : O/I; 9/21/25 : O/A

Price Target History: 1/27/21 : 65000; 10/26/21 : 55000; 3/18/22 : 50000; 8/9/22 : 43000; 1/12/23 : 35000; 2/21/24 : 34000; 3/28/24 : 48000; 4/3/25 : 25000; 8/8/25 : 48000; 10/21/25 : 72000; 3/23/26 : 160000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

A member of Research who had or could have had access to the research prior to completion owns securities (or related derivatives) in the Micron Technology Inc., NVIDIA Corp., Western Digital. This person is not a research analyst or a member of research analyst's household.

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory

(Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of FIT Hon Teng Ltd, Hon Hai Precision, Lenovo listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: <http://www.hkex.com.hk>.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The following companies do business in countries which are generally subject to comprehensive sanctions programs administered or enforced by the U.S. Department of the Treasury's Office of Foreign Assets Control ("OFAC") and by other countries and multi-national bodies: Samsung Electronics.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

© 2026 Morgan Stanley